EV Battery Course — Lesson 1.2: Electrical Fundamentals for EVs

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Module 1 · Fundamentals of Electric Vehicles and Batteries. This lesson gives you the electrical vocabulary every later module depends on: voltage, current, resistance, power, energy, Ah vs kWh, and series vs parallel.

1. Learning objectives

- Explain voltage, current and resistance using a mechanical analogy you already know.
- Use Ohm's law and the power equation to size currents and losses.
- Tell power (kW) from energy (kWh), and Ah from kWh.
- Calculate the voltage, capacity and energy of cells connected in series and parallel.
- Explain why EVs moved from 400 V toward 800 V.
- Explain why a pack needs hundreds or thousands of cells working together.

2. Fundamentals: the hydraulic analogy

Think of a hydraulic circuit on a press line.

Electrical quantity	Symbol, unit	Hydraulic equivalent	What it means physically
Voltage	V, volt	Pressure difference	The push that drives charge round a circuit
Current	I, ampere (A)	Flow rate (L/min)	Charge passing a point per second; 1 A = 1 coulomb/s
Resistance	R, ohm (Ω)	Pipe restriction	Opposition to flow; turns energy into heat
Power	P, watt (W)	Pressure × flow = hydraulic power	Rate of doing work
Energy	E, watt-hour (Wh)	Oil in the accumulator	Total work available
Charge capacity	Ah	Volume of the accumulator	How much charge can be delivered

The analogy breaks down in one place: electric current needs a closed loop. Electrons leave the negative terminal, do work in the load, and return to the positive terminal.

AC vs DC. Direct current (DC) flows one way at a steady voltage; batteries, solar panels and the EV's high-voltage bus are DC. Alternating current (AC) reverses direction many times a second (50 Hz in India); the grid and the traction motor's windings use AC. The onboard charger converts AC to DC, and the inverter converts DC back to variable-frequency AC for the motor.

3. Detailed technical explanation

Voltage is energy per unit charge: 1 V = 1 joule per coulomb. A lithium-ion cell's voltage comes from the difference in chemical potential of lithium in the anode and the cathode (Module 2). That sets the cell voltage: about 3.2 V nominal for LFP and 3.6–3.7 V for NMC and NCA.

Current is the flow of charge. Inside the wires it is electrons. Inside the cell it is lithium ions crossing the electrolyte. The two flows always balance: one electron through the external circuit for every Li^+ ion through the cell.

Resistance turns electrical energy into heat. In a pack it lives in the cells themselves (internal resistance), the busbars, the welds, the connectors and the fuses. Every milliohm matters at EV currents.

Power vs energy. Power is how fast energy moves; energy is how much there is. A pack's energy (kWh) sets range. Its power (kW) sets acceleration and charging speed. A 60 kWh pack delivering 150 kW is emptying itself at a rate that would drain it in 24 minutes.

Ah vs kWh. Ampere-hours measure charge, not energy. A 100 Ah cell at 3.2 V holds 320 Wh; a 100 Ah cell at 3.7 V holds 370 Wh. Always convert to Wh or kWh before comparing cells or packs of different chemistries.

Efficiency is useful output divided by input. Electrical losses are almost entirely I^2R heating plus switching losses in power electronics.

4. Diagram: series and parallel

```
flowchart LR
  subgraph S["Series: 3 cells, 3.2 V, 100 Ah each"]
    A1["Cell 1"] --> A2["Cell 2"] --> A3["Cell 3"]
  end
  subgraph P["Parallel: 3 cells, 3.2 V, 100 Ah each"]
    B1["Cell 1"]
    B2["Cell 2"]
    B3["Cell 3"]
  end
  S --> RS["9.6 V · 100 Ah · 960 Wh"]
  P --> RP["3.2 V · 300 Ah · 960 Wh"]
```

Series adds voltage; parallel adds capacity. Energy adds either way. A pack is written as **xSyP**: x cells in series, y in parallel.

5. Step-by-step working principle: what happens when you press the accelerator

1. The motor controller requests, say, 100 kW.
2. The pack voltage is about 350 V, so the BMS must allow $I = P \div V \approx 286$ A.
3. That current flows through every series group. With 2 cells in parallel, each cell carries about 143 A.
4. Each cell's internal resistance drops its terminal voltage ($V = I \times R$) and creates heat (I^2R).
5. The pack voltage sags under load, so even more current is needed for the same power.
6. The BMS watches the lowest cell voltage. If any cell approaches its minimum, it limits power for the whole pack.

The weakest cell limits the whole pack. That is why cell matching (Module 12) and balancing (Module 7) matter.

6. Engineering equations

$$V = I \times R \qquad \text{(Ohm's law)}$$

$$P = V \times I = I^2 R = \frac{V^2}{R}$$

$$E = P \times t \qquad E_{cell}\ (\text{Wh}) = V_{nom} \times C\ (\text{Ah})$$

$$V_{pack} = N_s \times V_{cell} \qquad C_{pack} = N_p \times C_{cell} \qquad E_{pack} = N_s \times N_p \times E_{cell}$$

$$\text{C-rate} = \frac{I\ (\text{A})}{C\ (\text{Ah})}$$

V in volts, I in amperes, R in ohms, P in watts, t in hours, E in watt-hours, C in ampere-hours, N_s cells in series, N_p cells in parallel. A 1C current empties a full cell in about one hour; 2C in about 30 minutes.

7. Worked examples

Example 1: cell energy. An LFP prismatic cell is rated 3.2 V, 100 Ah.

- E = 3.2 × 100 = **320 Wh**

Example 2: building a pack, 104S1P of that cell.

- V_pack = 104 × 3.2 = **332.8 V**
- C_pack = 1 × 100 = **100 Ah**
- E_pack = 332.8 × 100 = 33,280 Wh = **33.3 kWh**

Example 3: the same energy from a different cell. An NMC cell is 3.6 V, 50 Ah, arranged 96S2P.

- V_pack = 96 × 3.6 = **345.6 V**; C_pack = 2 × 50 = **100 Ah**
- E_pack = 345.6 × 100 = **34.6 kWh**, from 192 cells instead of 104

Example 4: why 800 V. A car draws 100 kW. Assume the same 0.1 Ω total path resistance at both voltages (a simplification: in reality resistance also changes with cell count and cable size).

Pack voltage	Current I = P ÷ V	Loss I^2R
350 V	285.7 A	$285.7^2 \times 0.1 \approx$ **8.2 kW**
800 V	125.0 A	$125^2 \times 0.1 \approx$ **1.6 kW**

Doubling voltage roughly halves current and quarters the I^2R loss in the same conductor. In practice engineers take the benefit as thinner cables, lighter busbars and much faster DC charging rather than as pure efficiency.

Example 5: C-rate. A 100 Ah cell supplies 150 A. C-rate = 150 ÷ 100 = **1.5C**.

8. Real-world EV examples

- **400 V class:** Most Indian EVs today, including the Tata Nexon EV and MG ZS EV, use packs around 300–400 V.
- **800 V class:** Hyundai's E-GMP platform (Ioniq 5, Kia EV6) and Porsche Taycan. 800 V lets them accept DC charging at well over 200 kW.
- **Two-wheelers:** Scooters run around 48–80 V. That stays near or below the 60 V DC threshold, which simplifies safety requirements.
- **Cell counts:** A cylindrical-cell pack can contain thousands of cells (Tesla packs using 2170 cells have several thousand). A large-prismatic LFP pack may have about 100–200 cells.

9. Manufacturing applications

- **Weld and joint resistance.** At 300 A, a weld with 0.5 mΩ extra resistance dissipates $300^2 \times 0.0005 = 45$ W. That is a local hot spot. Weld resistance is a critical-to-quality characteristic, checked by resistance measurement at end-of-line.
- **Cell matching.** In parallel groups, a low-resistance cell takes more than its share of current and ages faster. Cells are graded by capacity and internal resistance before assembly.
- **HV safety.** Above 60 V DC, plants need insulated tooling, interlocked stations, trained and authorised workers, and insulation-resistance tests.
- **Costing.** Cell price is quoted per kWh, not per Ah. Busbar copper cross-section is sized for current, so higher-voltage packs save copper.

10. Common mistakes and misconceptions

- **Comparing cells by Ah.** Always compare Wh; chemistries have different voltages.

- **Confusing kW and kWh.** A "60 kW battery" is not a thing; a 60 kWh battery delivering 60 kW is.
- **Assuming series cells share current unevenly.** In series every cell carries exactly the same current. Uneven sharing happens only in parallel.
- **Thinking 800 V is "more powerful."** Power depends on V × I. 800 V delivers the same power with less current.
- **Ignoring voltage sag.** Under load the terminal voltage drops, so current must rise to hold power.

11. Practical case study: hot busbar joints on a pack line

Problem. Thermal imaging during end-of-line discharge tests finds a few modules with one busbar joint running 15–20 °C hotter than its neighbours.

Analysis. Hot spots at a joint mean higher local resistance. Root causes found: weld penetration below spec on a laser-weld station after a lens change, plus surface oxide on some aluminium busbars from extended storage.

Corrective action. Add a post-weld joint-resistance check with a control limit, add a lens-cleaning and power-verification step at each shift start, and set FIFO and maximum storage time for busbars.

Lesson. At EV currents, an extra fraction of a milliohm is a reliability and safety issue, not a cosmetic one.

12. Revision notes

- $V = IR$. $P = VI = I^2R$. $E = Pt$.
- Series adds voltage; parallel adds capacity; energy adds either way.
- Ah is charge; Wh is energy. Convert before comparing.
- kW is the rate; kWh is the amount.
- Higher voltage means lower current, thinner conductors, lower I^2R and faster charging.
- The weakest cell limits the pack.

13. Glossary

Term	Meaning
Nominal voltage	Average voltage over a discharge; used to calculate energy

Term	Meaning
xSyP	x cells in series, y in parallel
C-rate	Current as a multiple of capacity
Internal resistance	The cell's own resistance; causes sag and heat
Voltage sag	Drop in terminal voltage under load
HV	High voltage: above 60 V DC in automotive practice
Busbar	Rigid conductor linking cells or modules

14. Knowledge check

Conceptual

1. Why do two 100 Ah cells of different chemistries not necessarily store the same energy?
2. In a 2P group, why does a low-resistance cell age faster?
3. Why does pack voltage sag under acceleration?
4. What changes, and what stays the same, when you move from a 400 V to an 800 V pack of the same energy?
5. Why does the BMS limit power based on the lowest cell, not the average?

Numerical

1. A cell is 3.7 V, 5 Ah. How many Wh? How many are needed for a 75 kWh pack?
2. A pack is 108S3P of 3.65 V, 60 Ah cells. Find V_pack, C_pack and E_pack.
3. A cell has 1.2 mΩ internal resistance and carries 180 A. Find the voltage drop and heat generated.

Application

1. You are costing two packs of equal kWh: 400 V and 800 V. Which cost lines change, and in which direction?
2. A scooter maker wants to move from 48 V to 72 V. What benefits and new requirements should they expect?

15. Answer key

Conceptual

1. Energy = voltage × Ah. Different chemistries have different voltages (LFP 3.2 V vs NMC about 3.7 V).
2. In parallel the voltage is common, so current divides inversely with resistance. The low-resistance cell carries more current, runs hotter and cycles harder.
3. Current through each cell's internal resistance creates a drop of I × R, and electrochemical polarisation adds more (Module 2).
4. Energy stays the same. Cells in series roughly double, cells in parallel halve, current halves, cables and busbars can be thinner, and charging power can rise. Insulation, connectors and power semiconductors need higher voltage ratings.
5. All cells in a series string carry the same current. If the lowest cell hits its voltage floor, discharging further would over-discharge and damage it, even if the others have energy left.

Numerical

1. 3.7 × 5 = **18.5 Wh**. 75,000 ÷ 18.5 = 4,054, so about **4,054 cells** (in practice rounded to a convenient xSyP arrangement).
2. V = 108 × 3.65 = **394.2 V**. C = 3 × 60 = **180 Ah**. E = 394.2 × 180 = 70,956 Wh ≈ **71.0 kWh**.
3. ΔV = 180 × 0.0012 = **0.216 V**. Heat = 180^2 × 0.0012 = **38.9 W** per cell.

Application

1. Cells per kWh are unchanged. Copper in busbars and HV cables goes down. Contactors, fuses, connectors, inverter semiconductors (often SiC) and insulation-rated parts go up. BMS channel count rises with more series cells. The net result depends on volume and supplier base.
2. Benefits: for the same power, current falls by a third, so lower losses, thinner cables and cooler connectors. New requirements: 72 V nominal can reach about 80 V or more when full, which is above 60 V DC. That brings stricter touch-protection, insulation and labelling requirements, plus a re-rated motor controller and charger.

16. Further learning

- *Electric Vehicle Technology Explained*, Larminie & Lowry: chapters on batteries and electrical machines.
- NPTEL, "Basic Electrical Technology" and "Introduction to Hybrid and Electric Vehicles": free video lectures.
- *Battery Management Systems for Large Lithium-Ion Battery Packs*, Davide Andrea (Artech House): practical, pack-level view.
- AIS-038 Rev 2: Indian requirements for EV electrical safety.

Next: Module 2, electrochemistry from first principles.